

RESEARCH ARTICLE

Sub-10 nm Silicon-Based Pyramidal Nanopore Enabling High Performance Single Protein Molecule Detection

Liang Cheng¹ | Chuhaio Cai¹ | Zhishan Yuan² | Li Ma¹ | Maoxiang Hou¹ | Huilong Liu¹ | Yu Xiao¹ | Mengjie Zheng¹ | Yun Chen¹ | Xin Chen¹ | Ching-Ping Wong³

¹State Key Laboratory of Precision Electronic Manufacturing Technology and Equipment, School of Electromechanical Engineering, Guangdong University of Technology, Guangzhou, China | ²State Key Laboratory of High Performance Tools, School of Electromechanical Engineering, Guangdong University of Technology, Guangzhou, China | ³Department of Electronic Engineering, The Chinese University of Hong Kong, Hong Kong, China

Correspondence: Yun Chen (chenyun@gdut.edu.cn)

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ABSTRACT

Solid-state nanopores are pivotal for next-generation single-protein detection, yet reconciling high fabrication efficiency, precision, and signal quality remains challenging. In this study, we integrate pyramidal silicon nanopores—valued for electric field focusing and high capture efficiency—with a silicon dioxide (SiO₂) layer designed to minimize nonspecific protein adsorption. This strategy aims to balance rapid translocation rates with clear blockade signals. Furthermore, we have developed a large-scale fabrication method capable of producing uniformly arranged silicon-based pyramidal nanopores with diameters below 10 nm. This technique combines wet etching, real-time ionic current monitoring, and thermal oxidation, enabling precise size control down to 7 nm. We evaluated 7 nm, 12 nm, and 14 nm nanopores for detecting α -synuclein (α -syn), immunoglobulin G (IgG), and ferritin (FTN), respectively. All nanopore sizes achieved efficient target detection, yielding resolvable blockade signals alongside high capture rates and fast translocation speeds. Notably, α -syn detection demonstrated improved overall performance in terms of event rate, signal quality, and blockade signal resolution, with similarly effective results observed for IgG and FTN. These results demonstrate the strong potential of this nanopore platform for protein detection.

1 | Introduction

Solid-state nanopores have emerged as a pivotal platform for next-generation protein molecule detection due to their label-free and high-throughput single-molecule sensing capabilities [1, 2]. Nevertheless, conventional solid-state nanopores, such as silicon nitride nanopores, as well as biological nanopores, exhibit inherent limitations in protein detection applications. Specifically, their symmetric cylindrical pore geometries induce electric field dispersion, which significantly prolongs molecular translocation time and consequently reduces detection efficiency [3, 4]. Although a moderate increase in translocation time can, within certain limits, improve the clarity of protein translocation

signals, the relatively hydrophobic nature of conventional solid-state nanopore surfaces tends to promote nonspecific protein adsorption. Moreover, extended translocation durations increase the time available for protein–surface interactions, thereby further aggravating nonspecific adsorption and resulting in noisy or ambiguous signal outputs, thus constituting a paradoxical challenge [5, 6]. Moreover, although biological nanopores offer single-molecule sensitivity, their limited mechanical stability and fixed pore sizes constrain their applicability for detecting larger biomolecules, including α -syn, IgG, and FTN [7–9].

To achieve clear protein detection signals at relatively rapid translocation speeds, silicon (Si) has been proposed as a highly

promising substrate material for solid-state nanopores [10–12]. Its anisotropic etching capabilities enable the fabrication of pyramidal nanopore structures, which contrast with the conventional cylindrical pores [13]. For instance, Wang et al. fabricated single nanopore smaller than 30 nm on silicon using comparable processing methods and confirmed their characteristic pyramidal internal morphology via focused ion beam (FIB) analysis [14]. Similarly, Hong et al. successfully fabricated nanopores averaging 15 nm in diameter on silicon substrates via wet etching techniques [15]. Furthermore, silicon's oxidation properties allow the formation of a hydrophilic silicon dioxide (SiO_2) surface layer, which imparts hydrophilicity to the nanopore surface and facilitates nanoscale pore size reduction through volumetric expansion during oxidation [16–18]. While the classical Deal-Grove model effectively describes macroscopic silicon oxidation kinetics [19], its accuracy diminishes for oxide layers thinner than 30 nm. Consequently, a modified Langmuir model has been developed to characterize the ultra-thin oxide regimes, providing a crucial theoretical framework for elucidating the early-stage silicon oxidation mechanisms [20]. Building on this, Deng et al. successfully employed thermal oxidation to reduce the size of silicon nanopores sizes from 50–60 nm down to less than 20 nm while preserving their pyramidal internal morphology [21]. Collectively, silicon-based nanopores possess asymmetric channel geometries, controllable pore size reduction via oxidation, and an inherently hydrophilic SiO_2 surface, collectively offering solutions to the common limitations observed in traditional solid-state and biological nanopores, such as electric field dispersion and nonspecific adsorption. These attributes synergistically enhance both the detection efficiency and accuracy of protein molecules, particularly for large proteins (molecular weight > 10 kDa) [22, 23].

Nevertheless, realizing these advantages necessitates the fabrication of uniform silicon-based nanopore arrays with diameters below 10 nm, a process hindered by two primary challenges. First, the absence of real-time, precise control over the termination of wet etching complicates the stable regulation of the top pore diameter of pyramidal nanopores below 30 nm, thereby limiting the effectiveness of subsequent oxidation-driven pore size reduction and affecting the uniformity of the final pore dimensions. This renders the precise tuning of pore diameters below 10 nm especially challenging. Second, during thermal oxidation, variations in the curvature at the base of pyramidal nanopores alter oxygen diffusion pathways, further complicating uniform pore size reduction [24].

This study introduces a strategy to reconcile the conflicting demands of rapid translocation speed and high-resolution protein detection signals. Initially, finite element and molecular dynamics simulations were employed to validate the electric field focusing effect of pyramidal nanopores and the hydrophilic properties of the silicon dioxide surface layer. Based on these insights, we developed an innovative fabrication technique for large-scale production of extensive, highly uniform silicon-based pyramidal nanopore arrays with pore diameters below 10 nm. This approach integrates real-time ionic current monitoring during the wet etching process with thermal oxidation to achieve precise size control. The fabricated nanopores were successfully applied to detect α -syn, IgG, and FTN, demonstrating high capture rates, rapid translocation speeds, and clear blockade

signals. In particular, the detection of α -syn protein exhibited breakthroughs in detection rate, temporal resolution, and signal-to-noise ratio relative to previously reported studies, thereby confirming the superior performance of the developed nanopore system in protein detection.

2 | Results and Discussion

2.1 | Fabrication and Characterization of Silicon-Based Nanopores

2.1.1 | Nanopores Design and Fabricating Technology

Figure 1a illustrates the design of the silicon-based pyramidal nanopore, in which an asymmetric pyramidal geometry is combined with a hydrophilic surface to enhance protein detection performance. The asymmetric structure generates an electric-field focusing effect that improves protein capture efficiency and translocation signals. In this study, a wide-to-narrow translocation mode was adopted by adjusting the applied bias polarity so that proteins entered from the wide opening and moved toward the narrow constriction. This transport mode was chosen because the field-focusing effect near the pore neck increases local capture probability and promotes detectable translocation events. Meanwhile, the electric field decays rapidly beyond the constriction, reducing the electrophoretic driving force after proteins pass through the pore neck and facilitating clearer post-constriction signals. Thus, this structure is beneficial not only for improving capture efficiency but also for characterizing translocation dynamics. In addition, the nanopore surface is coated with a negatively charged hydrophilic SiO_2 layer, which reduces nonspecific adsorption. By adjusting the buffer pH so that the proteins also carry negative charges, an electrostatic repulsive force (F_r) is established between the proteins and the nanopore wall. This repulsion drives the proteins to migrate along the nanopore center, significantly reducing nonspecific interactions with the sidewall and further suppressing adsorption.

To elucidate the advantages of pyramidal nanopores relative to conventional cylindrical nanopores in terms of electric field distribution, a finite element simulation model was developed with the details shown in the experimental section. As depicted in Figure 1b, the current density within the cylindrical nanopore is uniformly distributed. In contrast, the current density inside the pyramidal nanopore displays a gradient, progressively increasing from the entrance toward the exit, demonstrating a pronounced electric field focusing effect at the exit region. Notably, the current density at the pyramidal nanopore's exit is substantially greater than that observed in the cylindrical nanopore. A quantitative analysis along the central axes (paths a1–a2 and b1–b2 for the pyramidal and cylindrical nanopore, respectively) shows that the current density in the pyramidal nanopore increases nonlinearly from $1.5 \times 10^6 \text{ A/m}^2$ at the entrance to $1.8 \times 10^7 \text{ A/m}^2$ at the exit (Figure 1c). Conversely, the current density within the cylindrical nanopore remains approximately constant at $4.0 \times 10^6 \text{ A/m}^2$, reflecting a uniform distribution. This confirms that the pyramidal nanopore exhibits a significant electric field focusing effect, with an output current density approximately 4.5 times that of the cylindrical nanopore.

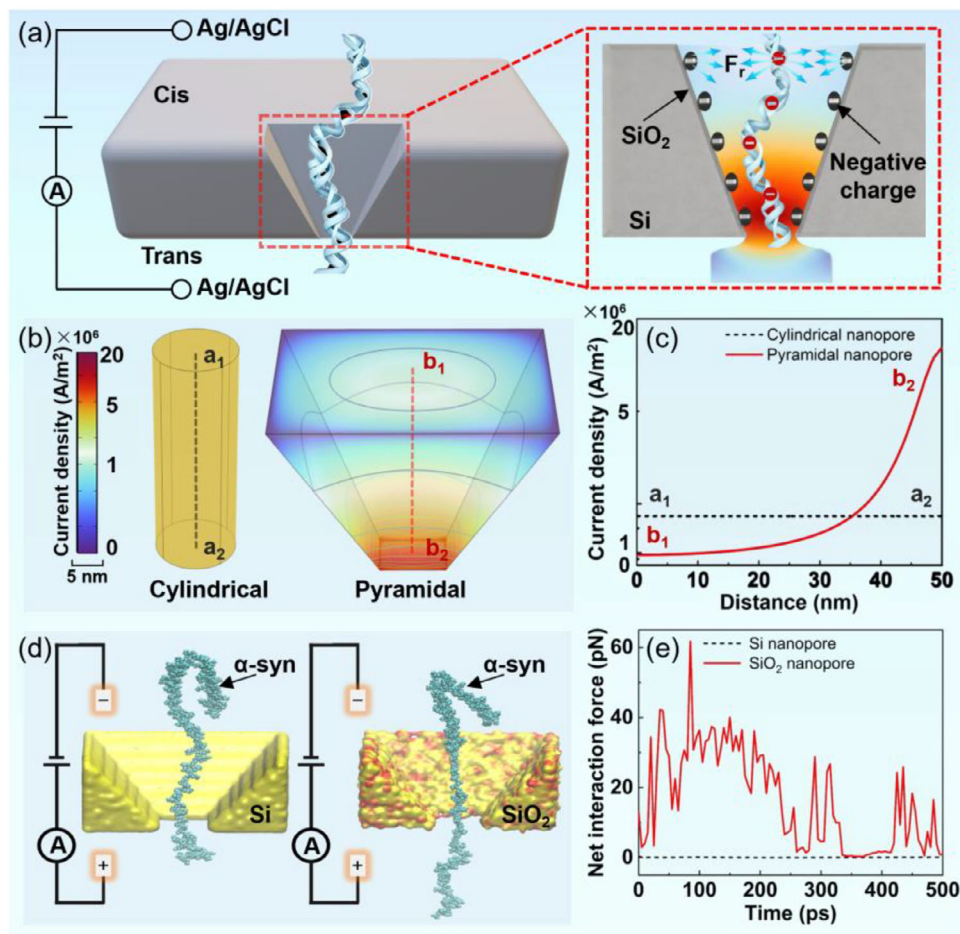


FIGURE 1 | Design of pyramidal nanopores. (a) Schematic of the design concept combining a pyramidal nanopore with a hydrophilic SiO₂ surface. (b) Electric field distribution contour maps of cylindrical and pyramidal nanopores; (c) Variation of current density along the central axis of each nanopore; (d) Snapshots from molecular dynamics simulations of α -syn protein translocation through Si and SiO₂ substrates; and (e) Corresponding temporal variation of interaction forces on Si and SiO₂ substrates.

Following the confirmation of the electric field focusing effect in the pyramidal nanopore, molecular dynamics simulations using NAMD were conducted to investigate the differential behavior of silicon and silicon dioxide nanopores during protein translocation. The visualization results of α -syn protein translocation through these nanopores (Figure 1d) reveal that proteins traverse a greater distance within the silicon dioxide nanopore compared to the silicon nanopore within an identical duration of 500 ps, demonstrating a faster translocation velocity. The temporal evolution of the net interaction force (Figure 1e) indicates that the SiO₂ system exhibits relatively strong protein-wall interactions at the initial stage of translocation, and that this interaction gradually weakens over time. The force shown here represents the interaction force between the protein and the nanopore wall. At the early stage of translocation, negatively charged α -syn experiences strong interactions with the negatively charged SiO₂ pore wall under electrostatic effects. As translocation proceeds, electrostatic repulsion drives the protein away from the pore wall and toward the center of the pore, thereby weakening the direct interaction between the protein and the pore wall. When the molecule approaches the central region of the nanopore, the interactions from different directions become more balanced, and the net interaction force is further reduced and approaches zero, thereby helping to reduce nonspecific adsorption during

translocation. In contrast, the electrically neutral Si surface cannot provide the strong electrostatic repulsive regulation present in SiO₂. As a result, the protein more readily approaches the pore wall, leading to nonspecific contact or even adsorption, which in turn reduces the translocation rate and gives rise to more irregular blockade signals. These simulation results further confirm the feasibility and advantages of truncated-pyramidal oxidized nanopores for protein detection.

2.1.2 | Fabrication of Nanopore Arrays

Figure 2a presents a schematic overview of the procedure utilized to fabricate silicon-based pyramidal nanopore arrays using the homemade setup shown in Figure S1 of the Supporting Information. Initially, the silicon wafers are subjected to ultrasonic cleaning in acetone to ensure surface cleanliness (Step I). The fabrication process begins with the deposition of silicon nitride mask layers on both sides of the silicon wafer (Step II). Subsequently, anisotropic etching is performed on the front side to create pyramidal microcavities (Step III). To improve fabrication efficiency, the wafer backside is rapidly thinned (Step IV). The backside silicon is then further etched in KOH at a reduced rate under real-time ionic current monitoring of nanopore opening

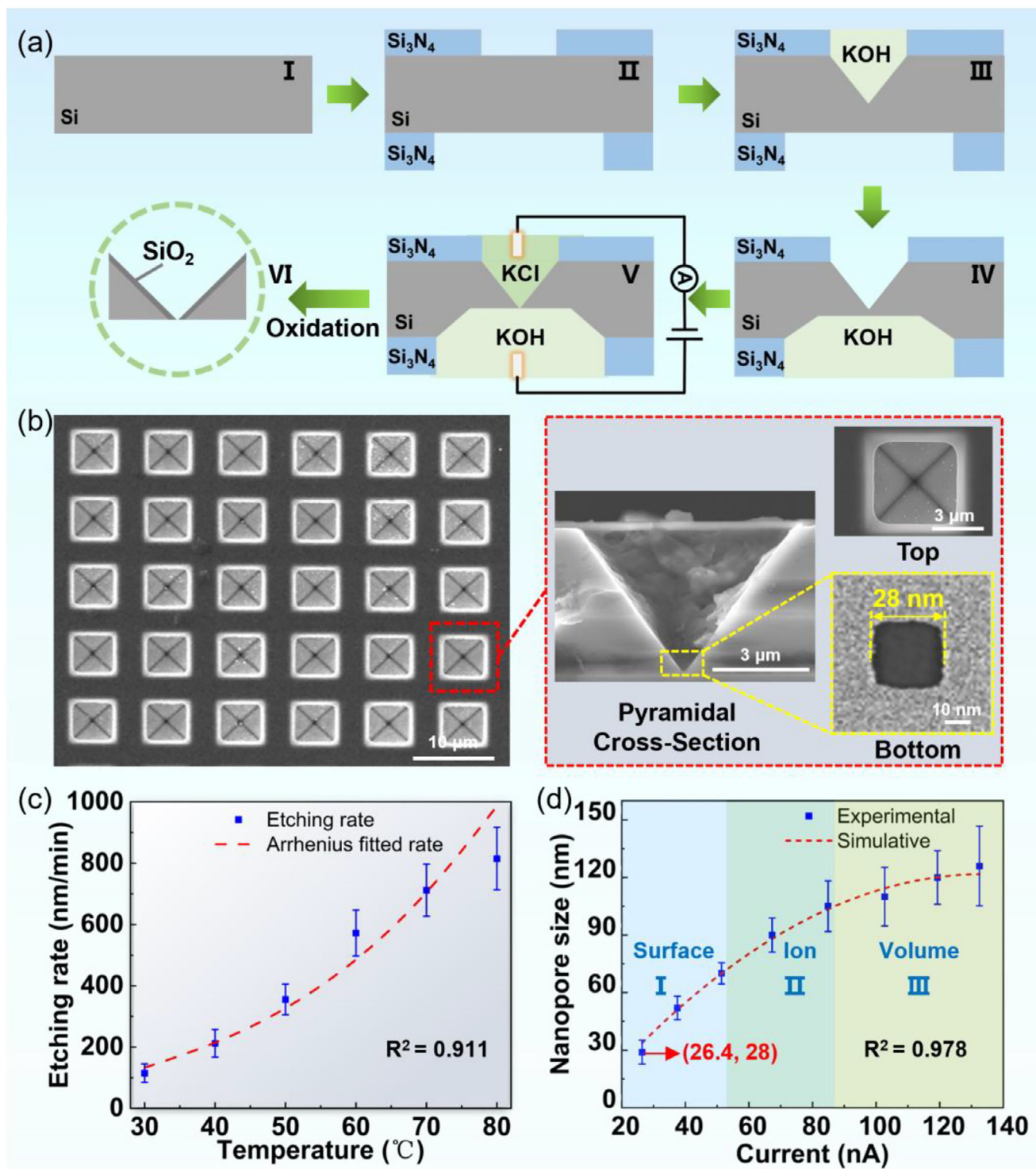


FIGURE 2 | Fabrication process and results of pyramidal nanopores. (a) Schematic of nanopore fabrication process; (b) Scanning electron microscopy (SEM) image of the silicon-based nanopore array; (c) Dependence of etching rate on temperature; and (d) Correlation between pore size and monitored ionic current, with a fitted curve based on the PNP model.

(Step V). A small external bias of 50–70 mV is applied to compensate for the initial interfacial electrostatic current and maintain the baseline current close to zero before pore opening. The reduced etching rate improves the uniformity of residual silicon thinning and facilitates synchronized pore opening across the 30-pore array. Once a significant current increase associated with through-hole formation is detected, the etching process

is immediately terminated. Finally, the fabricated nanopores undergo oxidation treatment in a tube furnace to complete the process (Step VI).

Figure 2b depicts the pyramidal microcavity structure formed via front-side etching in Step III, which preserves morphological integrity without exhibiting sidewall distortion due to

over-etching or profile deformation resulting from under-etching. To optimize the balance between processing efficiency and precision during the rapid backside thinning in Step IV, the effect of etching temperature on the thinning rate was systematically investigated, as shown in Figure 2c. The thinning rate of Si (100) increased with temperature and followed an Arrhenius-type relationship ($R^2 = 0.911$); the fitting methodology is provided in Figure S2 of the Supporting Information. Within the low-temperature range of 30°C–40°C, etching uniformity was maximized; however, the thinning rate remained below 220 nm min⁻¹, thereby limiting throughput. In contrast, at temperatures of 60°C or higher, the variability in thinning rate increased markedly, with fluctuations exceeding 70 nm min⁻¹, which increased the risk of over-etching. After comprehensive evaluation, an etching temperature of 50°C was selected as the optimal condition, providing a thinning rate of approximately 350 nm min⁻¹ with a variation limited to 50 nm min⁻¹, thereby ensuring a relatively uniform residual silicon thickness while maintaining satisfactory processing efficiency.

Following rapid thinning, a residual silicon thickness of 3–5 μm remains atop the tapered microcavity. To fabricate nanopores with minimal diameters during Step V, it is essential to further reduce the thinning rate to prevent over-etching and enable immediate termination of etching upon pore formation. To this end, the quantitative relationship between monitored ionic current (*I*) and nanopore diameter (*d*) was examined for an array of nanopores, as shown in Figure 2d. The initial abrupt current change corresponds to a current of 26.4 nA, which correlates with an average pore diameter of 28 ± 6 nm. Notably, the ionic current exhibits a nonlinear relationship with the pore diameter at this stage, deviating significantly from classical Ohm's law; this behavior is accurately captured by a modified Poisson–Nernst–Planck (PNP) transport model with detailed derivations of the relevant equations and the fitting methodologies provided in Figure S3 of the Supporting Information. The experimental data exhibit excellent agreement with the modified model, yielding an R^2 value of 0.978.

The fitting results (Figure 2d) demonstrate that ionic currents ranging from 26.4 to 132.5 nA correspond to nanopore diameters between 28 and 126 nm. Detailed analysis of the fitting results reveals that the dominant transport mechanism varies across current intervals, influencing the relationship between pore diameter and ionic current. Specifically, in the low-current regime ($I < 50$ nA, Stage I), surface charge effects predominate, with the electroosmotic flow driving rapid pore expansion at a rate of approximately 2.1 nm/nA. In the intermediate current regime (50–90 nA, Stage II), enhanced ion crowding alters transport dynamics, significantly reducing the pore size growth rate. At high currents (> 90 nA, Stage III), volume conduction dominates, and the pore size increases at a slower rate of approximately 1.7 nm/nA.

It should be noted that, the analysis identifies two principal challenges in monitoring nanopore opening currents. First, in the low-current regime, strong electroosmotic flow driven by surface charges obscures weak initial opening signals within noise, complicating detection. Second, current superposition effects from pore clusters cause signal overlap, whereby the abrupt current generated by the first pore-opening event masks

subsequent weaker signals. The interplay of these factors results in the first detected effective abrupt current during pore opening (Step IV) reaching 26.4 nA, rather than a theoretically lower threshold [25].

2.1.3 | Thermal Oxidation-Induced Nanopore Shrinkage

To achieve smaller nanopore diameters, the samples were subjected to thermal oxidation treatment. As shown in the SEM images in Figure 3a, the nanopore diameter gradually decreased with increasing oxidation time, shrinking from an initial 28 to 7 nm after 8 h of oxidation. Figure 3b further quantifies this time-dependent shrinkage behavior. The experimental data were well fitted by a Langmuir-type model ($R^2 = 0.997$) with the fitting methodologies provided in Figure S4 of the Supporting Information, indicating a saturation-like decrease in pore diameter during the oxidation process. Consistently, the oxide layer thickness in Figure 3c also exhibited a saturation trend, increasing approximately linearly during the initial stage (1–3 h) and transitioning to near-parabolic growth at longer oxidation times (4–8 h).

To verify the formation of SiO₂, energy-dispersive X-ray spectroscopy (EDS) analysis was performed on the nanopore surfaces. The results, depicted in Figure 3e, show cross-sectional EDS line scans and area scans that clearly reveal elemental distribution characteristics: a uniform region with significant oxygen enrichment is observed above the silicon substrate surface. Correspondingly, the silicon signal intensity increases from the oxide layer toward the silicon substrate. The spatial coincidence of the oxygen-enriched region with the silicon intensity attenuation zone strongly confirms the successful growth of a SiO₂ layer on the silicon substrate surface. The oxide layer formed post-oxidation was characterized using XRD and XPS. The XRD spectrum (Figure 3d) exhibits a broadened envelope peak near 22° superimposed on the Si substrate background, confirming the presence of SiO₂. Although the (100) orientation of the Si wafer theoretically suppresses diffraction from the (400) crystal plane, slight wafer misalignment permits diffraction, resulting in a sharp Si (400) peak indicative of substrate detection. The total XPS spectrum (Figure 3f) reveals prominent Si and O core-level peaks without significant impurity, confirming that the oxide layer's purity and composition. High-resolution Si 2p spectrum (Figure 3g) displays characteristic double peaks, with the low binding energy peak at 99.5–100 eV attributed to the basal Si⁰, while the high binding energy peak at 103.5–104 eV attributed to the Si⁴⁺ in SiO₂. The high-resolution spectrum of the O 1s (Figure 3h) shows that a dominant peak at 533 eV, consistent with Si–O bonding and confirming the formation of SiO₂.

2.2 | Evaluation of Translocation Performance in Silicon-based Nanopores

2.2.1 | Protein Detection Principles and Setup

To evaluate the efficacy of the fabricated nanopores in protein detection, a nanopore detection system was established, the operational principle of which is illustrated in Figure 4a. When proteins translocate through the nanopore, they modulate the

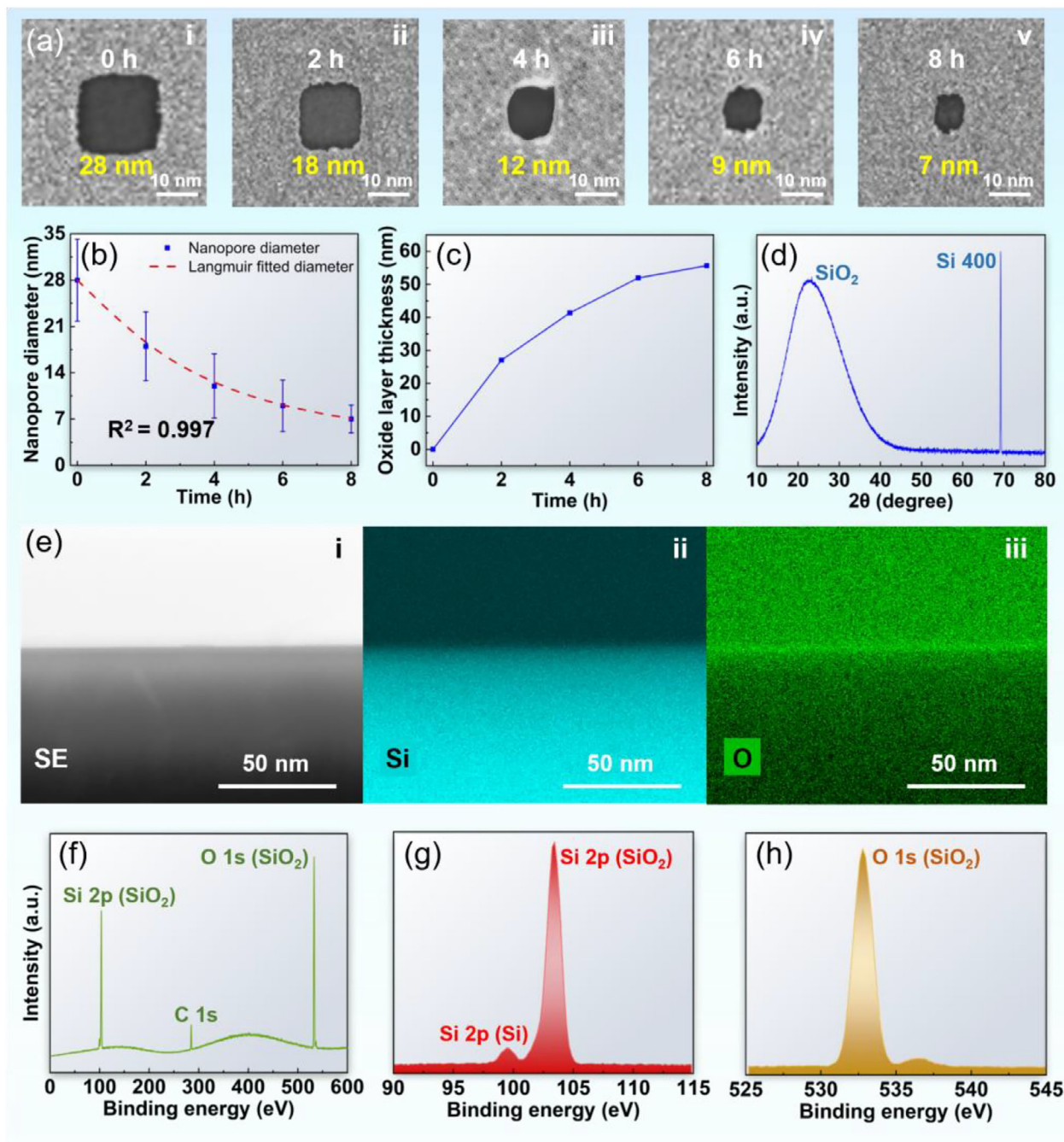


FIGURE 3 | Thermal oxidation-induced nanopore shrinkage results. (a) SEM images illustrating pore size reduction over time; (b) Temporal evolution of nanopore diameter; (c) Temporal evolution of oxide thickness; (d) X-ray diffraction (XRD) characterization of the oxidized sample; (e) Secondary electron image of SiO₂ and corresponding elemental distribution maps; (f) Total X-ray photoelectron spectroscopy (XPS) spectrum; (g) High-resolution Si 2p spectrum; and (h) High-resolution O 1s spectrum.

baseline ionic current, generating characteristic blockade current signals. This process is characterized by two key parameters: the change in blockade current (ΔI) and the translocation duration (ΔT), which together define a translocation event. The core component of the system is a flow cell, wherein an electric field drives proteins from the cis chamber through the nanopore into the trans chamber. To maintain a continuous flow path through the nanopore connecting the two chambers, a sandwich structure was employed to separate the chambers, as depicted in Figure 4b. In this configuration, a silicon chip is clamped

between two O-rings, providing sealing and compartmentalization of the chambers. The complete protein detection setup is shown in Figure S5 of the supporting information. In addition, the I - V characteristics of the nanopore chips used for protein detection were repeatedly measured, and the corresponding curves are provided in Figure S6 of the Supporting Information. The measured I - V curves exhibited approximately linear behavior over the tested bias range, further confirming the stability of ionic transport through the nanopores and providing a reliable basis for subsequent protein detection.

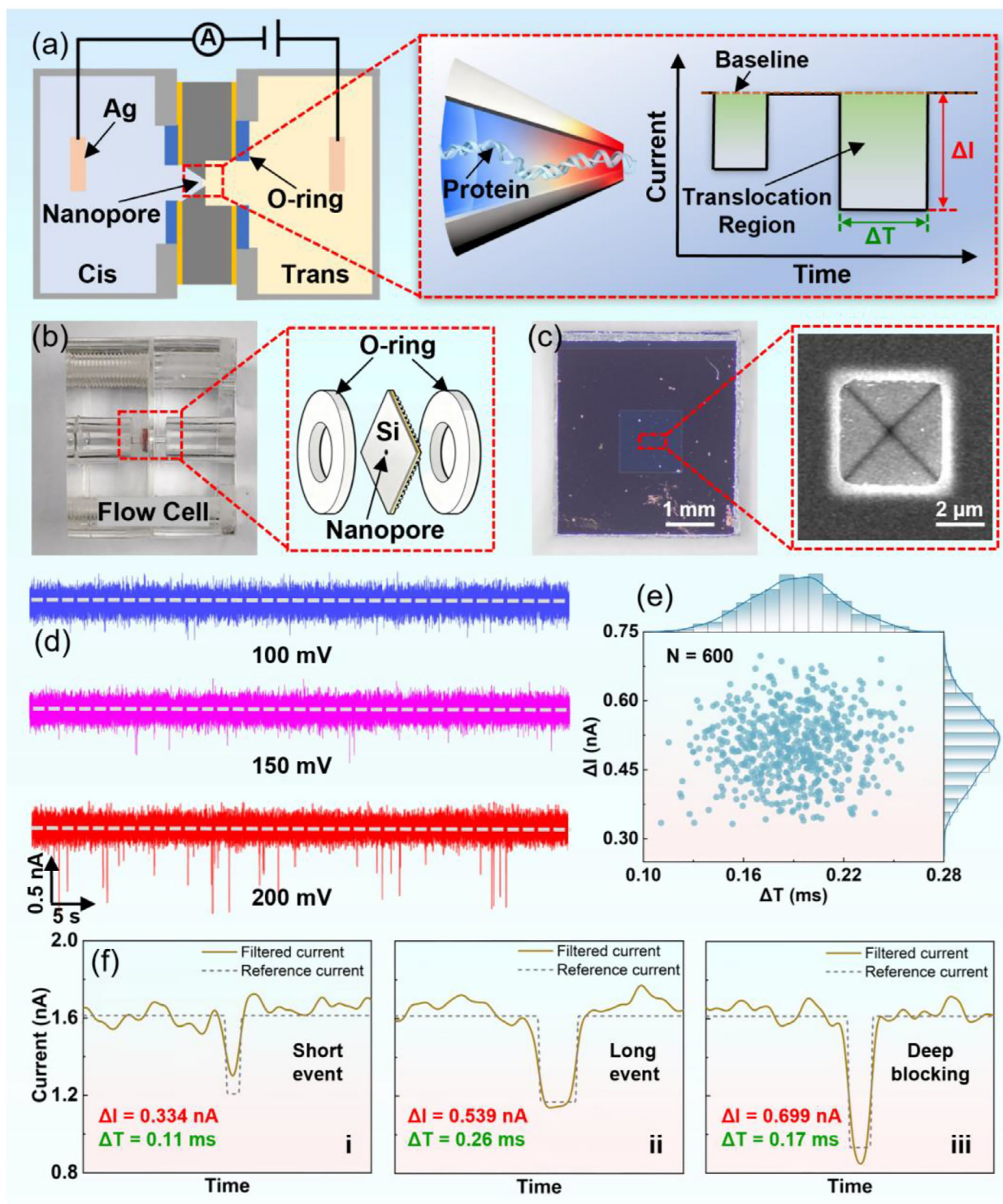


FIGURE 4 | Schematic Representation of Protein Detection and Analysis of α -syn Translocation. (a) Principle underlying protein detection; (b) Photograph and structural diagram of the flow cell; (c) Image of the protein detection chip; (d) Current blockade signals recorded under varying bias voltages; (e) Distribution of translocation events; and (f) Representative translocation event data of 200 mV.

Figure 4c shows a photograph of the protein detection chip, which incorporates a single pyramidal oxide nanopore within the silicon substrate. To prove the universal capability of the translocation behavior of macromolecular proteins through the fabricated silicon-based nanopores, detection experiments were performed including three kinds of proteins, including α -syn (with a diameter of 5 nm), IgG (with a diameter of 10 nm),

and FTN (with a diameter of 12 nm), which are large, complex biomolecules critical to various biological processes. Due to their significant size and complex folding, they cannot be represented by a simplified molecular-input line-entry system (SMILES) string, which is typically used for small organic molecules. Their structures are instead defined by their amino acid sequences and complex three-dimensional conformations. Therefore, although

TABLE 1 | Comparison of protein detection performance.

Nanopore type	Event rate (events s ⁻¹)	Translocation duration (ms)	Signal-to-noise ratio (S/N)
Silicon Nitride [26]	0.3–0.5	1–5	3.1–3.6
Biological [7]	0.17	22	2.3
2D Materials [27]	<0.2	<0.1	4.0
Silicon-based (This work)	1.57	0.18	5.6

the front face of the pyramidal pore has a fixed edge length of 4 μm , the nanopore diameter at the apex is customized according to the size of the target protein.

Figure 4d presents the blockade current signals corresponding to the detection of α -syn using a nanopore with a tip diameter of 7 nm. The data indicate that characteristic blockade current signals associated with α -syn translocation were observed under all applied bias voltages. However, at 100 mV, the blockade current amplitude was relatively weak and the signal-to-noise ratio was low, suggesting that this voltage was insufficient to overcome the electrostatic repulsion exerted by the negatively charged α -syn protein, resulting in a limited number of detected translocation events. At 150 and 200 mV, the amplitude of the blockade current increased significantly, accompanied by a marked improvement in protein detection rates, with detailed results provided in Figure S7 of the Supporting Information. Notably, at 200 mV, the distributions of dwell time (ΔT) and current blockade amplitude (ΔI) were statistically analyzed based on an expanded dataset containing approximately 600 translocation events, as shown in Figure 4e. The ΔT values ranged from 0.11 to 0.26 ms, while ΔI values spanned from 0.31 to 0.69 nA, with mean values of 0.18 ms and 0.56 nA, respectively. The event rate was 1.57 events s⁻¹. Both ΔT and ΔI exhibited normal distributions, indicating that the dominant event population exhibited consistent translocation behavior under the applied electric field. Furthermore, three representative event classes—short events, long events, and deep blockade events—were selected for detailed analysis, as shown in Figure 4f. These events predominantly exhibited single-peak blockade characteristics, demonstrating that the detected signals were dominated by translocation-related processes. Short events represent rapid and direct protein passage through the pore constriction, with short residence time and minimal pore-wall interaction. Long events represent translocation with extended residence in the sensing region, reflecting configurational adjustment, rotational reorientation, or transient interaction with the pore wall or constriction during passage. Deep blockade events represent protein states that produce a stronger current blockade because a larger effective excluded volume is formed in the sensing region under a specific orientation or transient conformation. This behavior agrees with the physical characteristics of the nanopore system, namely the electric-field focusing effect of the pyramidal structure and the hydrophilic SiO₂ surface, which together enhance capture efficiency while suppressing excessive nonspecific interactions. In this work, these event classes are discussed as mechanistically distinct translocation signatures derived from ionic current analysis.

A comparative analysis was performed between the α -syn detection performance of the silicon-based nanopores developed in this study and that of other nanopore materials reported in the literature (Table 1). The data indicate that the event rate achieved here, 1.57 events s⁻¹, is substantially higher than those reported for silicon nitride nanopores [26] (0.3–0.5 events s⁻¹) and biological nanopores [7] (0.17 events s⁻¹). The mean translocation duration of 0.18 ms falls within a relatively short yet still resolvable time window for the present recording system. Compared with extremely fast events (<0.1 ms) reported for some two-dimensional (2D) material nanopores [27], the translocation signals observed here are more readily distinguished and remain associated with clear blockade features. In contrast to the much longer translocation durations (1–5 ms) often reported for silicon nitride nanopores, the present platform maintains a higher event rate while preserving a high signal-to-noise ratio. In terms of signal quality, the silicon-based nanopores achieved a signal-to-noise ratio of 5.6, markedly higher than those reported for silicon nitride (3.1–3.6), biological (2.3), and two-dimensional material nanopores (4.0). Collectively, these findings indicate that the silicon-based nanopores developed in this work provide a favorable combination of event frequency, signal resolvability, and signal quality for α -syn protein detection.

2.2.2 | Detection of IgG and F₁TN

Moreover, Figure 5a presents the blockade current signals recorded during the detection of IgG using a nanopore with a tip diameter of 12 nm. Distinct blockade signals corresponding to IgG translocation were observed under three different applied bias voltages. Both the blockade current amplitude and protein detection rate increased with rising bias voltage, indicating that electrostatic repulsion between IgG molecules and the nanopore walls effectively reduced protein-wall interactions. This effect, combined with the inherent electric field focusing characteristic of the pyramidal nanopore, facilitated rapid and efficient translocation of IgG molecules. At 200 mV, the distributions of dwell time (ΔT) and current blockade amplitude (ΔI) were statistically analyzed using an enlarged dataset obtained from extended recording measurements, comprising approximately 600 translocation events, as depicted in Figure 5b. The ΔT values ranged from 0.22 to 0.52 ms, and ΔI values ranged from 1.15 to 2.07 nA, with mean values of 0.37 ms and 1.53 nA, respectively. The event rate was 1.01 events s⁻¹. Both ΔT and ΔI exhibited approximately normal distributions. Analysis of characteristic translocation events at 200 mV, shown in Figure 5c,

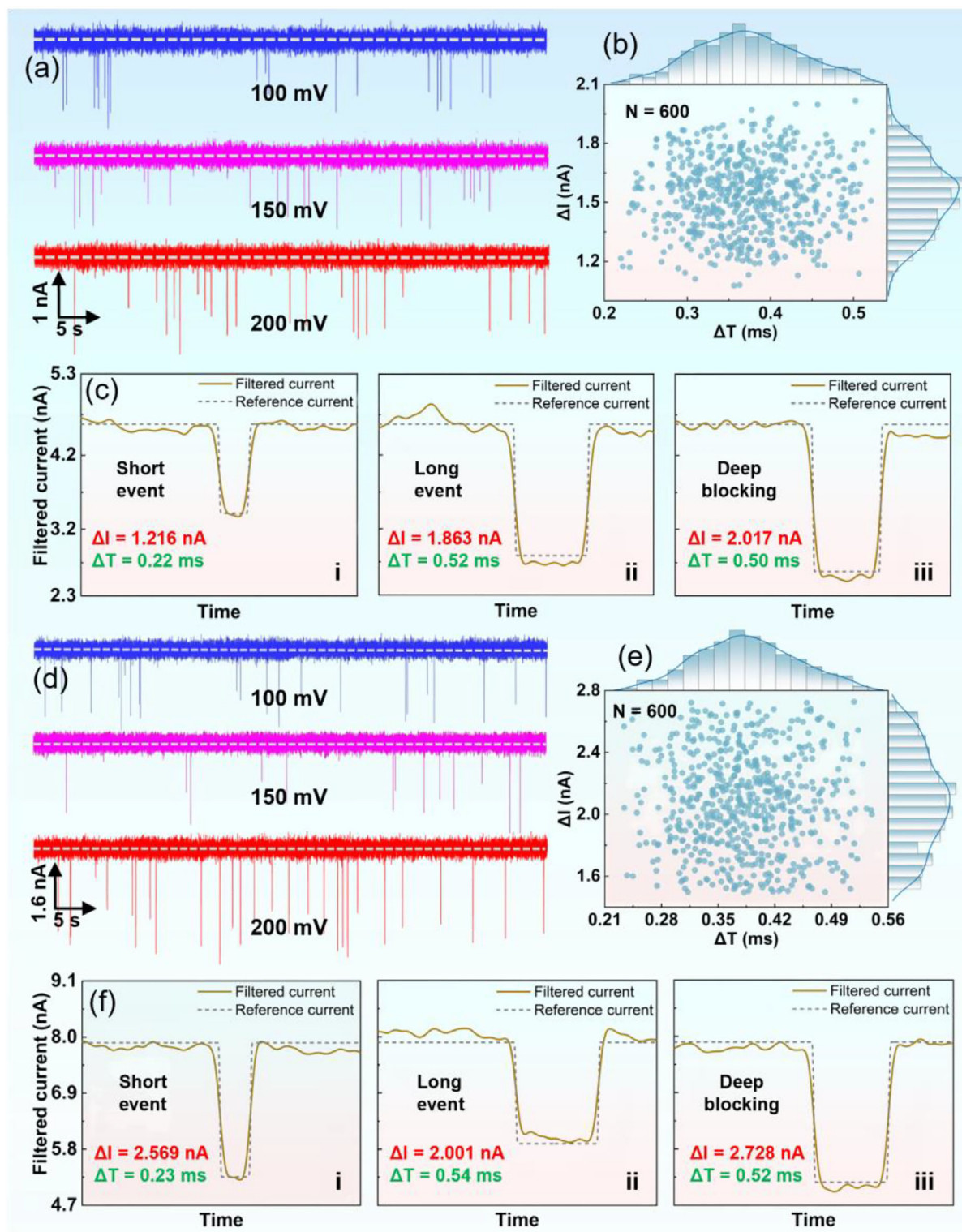


FIGURE 5 | Results of IgG and FTN protein translocation analysis. (a) Blockade current signals under different bias voltages (IgG); (b) Distribution of translocation events; (c) Typical event data of 200 mV; (d) Blockade current signals under different bias voltages (FTN); (e) Distribution of translocation events; and (f) Typical event data of 200 mV.

revealed distinct single-peak blockade signals, indicating that IgG proteins translocated through the nanopore in an unfolded yet intact state. These results suggest that the majority of IgG molecules were detected by the nanopore at a stable rate and conformation.

Furthermore, Figure 5e summarizes the detection results of FTN using a nanopore with a tip diameter of 14 nm, as illustrated in Figure 5d. The data confirm successful detection of FTN under various applied bias voltages. Specifically, Figure 5e,f demonstrate that at 200 mV, the distributions of dwell time (ΔT) and current

blockade amplitude (ΔI) for FTN translocation events exhibited normal distributions. Characteristic translocation events further confirmed that this nanopore enabled rapid and clear detection of FTN, with an average ΔT of 0.388 ms, an average ΔI of 2.087 nA, and an event rate of 1.00 events s^{-1} .

In summary, by tailoring the nanopore diameter, effective detection of proteins with varying sizes was achieved, validating the design objectives of the nanopore system developed herein.

3 | Conclusion

This study presents an integrated approach encompassing the design, fabrication, and application of silicon-based pyramidal nanopores specifically tailored for high-performance protein detection. Finite element and molecular dynamics simulations were employed to validate the advantages of the pyramidal geometry in enhancing electric field focusing, while the silica surface coating effectively mitigated nonspecific protein adsorption. Building upon these insights, a large-scale manufacturing strategy was developed, combining wet etching, real-time ionic current monitoring, and thermal oxidation. This method enables precise control over pore size, facilitating the production of highly uniform nanopore arrays with diameters below 10 nm, achieving a minimum pore size of 7 nm. The fabricated nanopores were successfully employed to detect three proteins of varying molecular sizes— α -syn, IgG, and FTN—demonstrating that tuning nanopore dimensions allows for selective analysis of proteins with different sizes. When benchmarked against previous studies, the silicon-based nanopore platform presented in this work shows a favorable combination of detection characteristics for α -syn, including a high event rate, good signal fidelity, and clear blockade resolution, while maintaining equivalent analytical performance for both IgG and FTN detection. The synergistic effect of the pyramidal structure and hydrophilic silica coating enhanced capture efficiency, accelerated translocation rates, and generated distinct high-fidelity blockade signals. Overall, this work addresses the longstanding challenge in conventional protein detection of balancing rapid translocation rates with clear blockade signals, establishing a robust and scalable platform for advanced, label-free, single-molecule protein analysis. This platform holds significant potential for applications in diagnostic technologies and fundamental biological research.

4 | Experimental Methodology and Simulation Framework

4.1 | Fabrication of Silicon-Based Pyramidal Nanopore Arrays

A p-type (100)-oriented single-crystal silicon wafer with a thickness of 200 μm was severed as the substrate. Silicon nitride masking layers of 100 nm thickness on the front side and 300 nm on the back side were deposited via chemical vapor deposition. Photolithography patterning was utilized to define square array etching windows arranged in a 5 μm pitch array with 4 μm side lengths on the front surface, as well as a single 5 mm square window on the rear surface. The front side was subsequently subjected to anisotropic etching in a 30% w/w KOH solution

maintained at 40°C to generate inverted pyramidal structures. Then the back side underwent rapid thinning in a 30% w/w KOH solution at 50°C, followed by a slower etching process at 25°C in the same solution. Ionic current signals were continuously monitored in real-time using an electrometer. Upon observing a pronounced current increase, indicative of via formation, the etching was immediately terminated. Finally, the sample containing the via was thermally oxidized in a tube furnace under a constant oxygen flow of 100 mL/min.

4.2 | Simulation Approaches and Model

4.2.1 | Simulation of Electrical Potential Distribution Within Nanopores

The electrical potential distribution inside nanopores was investigated using COMSOL Multiphysics. A three-dimensional (3D) model was constructed to compare two nanopore geometries: a pyramidal nanopore with a sidewall angle of 54°, an apex diameter of 10 nm, and a height of 50 nm, and a cylindrical nanopore with a diameter of 10 nm and the same height of 50 nm. The computational domain was discretized using physics-controlled meshing, with local mesh refinement in the nanopore region to improve numerical accuracy. The tip side of the nanopore was assigned a positive electric potential, whereas the bottom side was grounded, and all other boundaries were treated as electrically insulating boundaries. A potential difference of 200 mV was applied across the nanochannel to generate the electric field, allowing steady-state analysis of the spatial distribution and intensity of the electric field in the two nanopore geometries. A schematic of the simulation geometry and boundary conditions is provided in Figure S8 of the Supporting Information, and the main model settings are summarized in Table S1 of the Supporting Information. The simulation was performed to evaluate the electric-field focusing effect of the pyramidal nanopore under an applied bias.

4.2.2 | Molecular Dynamics Model for Protein Translocation

Protein translocation simulations were performed using the NAMD software package. Nanopore substrates composed of diamond-structured silicon and amorphous silicon dioxide, each measuring 129 Å $\times$ 129 Å $\times$ 58 Å, were constructed. A pyramidal through-pore was created at the center of each substrate, featuring a 7 nm edge length at the narrow end and a sidewall angle of 54°, consistent with the anisotropic wet etching characteristics of silicon. To ensure geometric comparability, both substrates shared identical pore geometries, while negative surface charges were introduced on the SiO₂ substrate to reflect its intrinsic surface properties.

An intrinsically disordered human α -syn protein was selected as the model molecule. The initial protein structure was obtained from the Protein Data Bank (PDB ID: 1XQ8), corresponding to the solution NMR structure of micelle-bound α -syn. Further details are provided in Table S2 of the Supporting Information. To ensure consistent initial configurations across different systems, all protein–nanopore models were constructed using the same

scripted alignment procedure in VMD. The protein was initially placed along the Z axis, with one end positioned at the narrow side of the nanopore, and the lowest atom of the protein was aligned to the nanopore center in the xy plane and to the bottom-layer height of the nanopore along z. Further details are provided in Figure S9 of the Supporting Information.

The protein–nanopore systems were solvated in 0.15 M NaCl aqueous solution, and an external electric field of 0.04 V/Å was applied along the Z-axis direction to drive protein migration through the pore axis. Atomic interactions were described using the CHARMM36 force field. The systems were first subjected to 10 000 steps of energy minimization using a conjugate gradient algorithm combined with a line-search Newton scheme to remove unfavorable structural contacts. Subsequently, 500 000 steps of equilibration were performed in the NPT ensemble using a Langevin piston barostat. Production simulations were then carried out for 4 ns in the NVT ensemble at 298 K using a Langevin thermostat, with a time step of 1 fs. The 500 ps trajectory shown in the manuscript represents a typical segment extracted from the full 4 ns production trajectory for visualization and interaction analysis. Throughout the simulations, system energy, pressure, atomic trajectories, and velocities were continuously recorded. The total interaction force between the protein and the nanopore was calculated using the built-in pairInteraction method in NAMD, based on the van der Waals and electrostatic contributions between the protein atoms and the nanopore atoms. Further details of the simulation setup and procedures are provided in Table S3 of the Supporting Information.

4.3 | Experimental Platform for Molecular Translocation

α -syn, IgG, and FTN were used as model proteins for nanopore translocation experiments. A three-dimensional acrylic liquid cell was used to form physically separated cis and trans chambers. Unless otherwise stated, both chambers contained the same electrolyte, consisting of 1 M KCl, 20 mM Tris-HCl, 1 mM MgCl₂, and 120 mM NaCl (pH 9.0). Protein samples were prepared in the same electrolyte. The concentrations of α -syn, IgG, and FTN were each 0.1 mg mL^{−1}. Detailed physicochemical information for α -syn, IgG, and FTN is summarized in Table S4 of the Supporting information. Under these conditions, the proteins were negatively charged. Bias voltages of 100, 150, or 200 mV were applied through Ag/AgCl electrodes using an electrical driving module integrated with a patch-clamp amplifier and a data acquisition system. Ionic current was recorded with an Axopatch 700B patch-clamp amplifier. The offset current was adjusted to zero before each measurement. A 10 kHz low-pass Bessel filter was applied during acquisition. Translocation events were identified from the blockade amplitude (ΔI) and dwell time (ΔT), and power spectral density analysis was performed in MATLAB. The entire measurement setup was enclosed in an aluminum-alloy Faraday cage to minimize electrical noise as shown in Figure S5.

4.4 | Nanopore Characterization

The dimensions and morphology of the nanopores were characterized using a ZEISS Sigma 300 field-emission SEM. EDS was

employed to analyze the elemental distribution of the oxidized nanopore surfaces. XRD patterns were acquired using a Rigaku SmartLab SE diffractometer. In addition, XPS analyses were performed using a Thermo Scientific K-Alpha spectrometer.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adfm76232-sup-0001-SupMat.docx.

Supporting File 2: adfm76232-sup-0002-MovieS1.gif.

Supporting File 3: adfm76232-sup-0003-MovieS2.gif.